

Clinical analysis of adverse drug reactions and pharmacovigilance

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Introduction

Drugs are developed for a specific purpose based on their pharmacology or drug action, to cure or control a disease or to aid in diagnosis of a disease. Although a drug may have a targeted effect, it may cause some undesirable effects during its clinical use. All drugs have risks. Throughout the drug development and approval process, adverse drug reactions (ADRs) are monitored and assessed to determine a benefit-risk profile for the drug. Even so, the risk profile of a drug is often not fully delineated at the time of regulatory approval as very rare adverse events may be undetected and the risk in vulnerable populations is often unknown, as was shown with chloramphenicol and development of blood dyscrasias in the 1950s [1]. As a result, ADR assessment, monitoring, and surveillance are necessary throughout a drug's entire life cycle in order to enhance our knowledge about its risks and safety. The term drug as used in this chapter refers to both human drugs and therapeutic biological products.

ADRs are an important public health issue that can cause significant morbidity and mortality. It has been estimated that approximately 5% of hospital admissions are related to ADRs and about 10%–20% of hospitalized patients experience an ADR while in the hospital [2–4]. The true incidence of ADRs may be higher because ADRs are often underreported; in addition, health care providers and patients may not recognize the relatedness of symptoms to a drug. A systematic review of data from 12 countries estimated underreporting of ADRs to spontaneous reporting systems to exceed 90% [5]. There is also some confusion when comparing incidence estimates because medical literature often groups ADRs with adverse drug events (ADEs) that may include medication errors, inappropriate use, or overdoses and may not be directly related to the mechanism of drug action [6].

ADRs are costly and involve the use of significant resources. ADRs may present as minor symptoms or cause serious harm or death and therefore should be identified early in the course of therapy to prevent further harm. Depending on the severity of the ADR, patients may require hospitalization or extension of hospitalization, significantly increasing health care expenditures. When ADRs are not recognized, additional medications may be added to treat symptoms and present added risks. Clinical diagnosis may not be apparent and require additional testing, procedures, or readmission. Besides direct costs, indirect costs may be incurred such as missed days of work or loss of employment [7]. The economic impact of an ADR varies depending on outpatient or inpatient setting, severity of the reaction, and length of time and resources necessary for appropriate diagnosis and treatment. Therefore health care providers need to have an adequate understanding of ADRs and consider an ADR as a potential cause of a patient's symptoms instead of attributing it to their underlying illness.

For newly approved drugs, our knowledge and understanding of ADRs is primarily based on the premarketing development program (i.e., animal, pharmacology, and clinical studies). Because of the limitations of these data sources, ADR knowledge continues to mature in the postmarketing setting through spontaneous reporting, observational studies, registries, and other information sources that reflect experience from broader utilization. *Pharmacovigilance*, as defined by the World Health Organization (WHO), includes the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other possible drug-related problems [8]. Pharmacovigilance efforts are necessary to ensure a well-informed risk profile. Unfortunately, many health care providers are not aware that drug labels are continuously updated with safety information as it arises and that their clinical experience in reporting ADRs is an important element of postmarket safety surveillance [9].

Definitions and classification

Definitions

The terms “adverse event” or “adverse drug event” are often used interchangeably with “adverse drug reaction”; however, these terms have different meanings. An *adverse event* is a broader term defined by the WHO as a “medical occurrence temporally associated with the use of a medicinal product, but not necessarily causally related.” Regulatory agencies such as the Food and Drug Administration (FDA) define an *adverse event* (also referred to as an *adverse experience*) as any untoward medical occurrence associated with the use of a drug in humans, whether or not considered drug related [10]. These definitions include an adverse event occurring under any circumstance in the course of a drug’s use in professional practice: from drug overdose whether accidental or intentional, from drug abuse, from drug withdrawal, or with any failure of expected pharmacological action, and do not require a causal relationship [11]. According to the WHO, an *adverse drug reaction* is a response to a drug that is noxious and unintended and which occurs at doses normally used for prophylaxis, diagnosis, or therapy of a disease or for modification of physiologic function [12]. Edwards and Aronson [13] further distinguished an ADR as “an appreciably harmful or unpleasant reaction, resulting from an intervention related to the use of a medicinal product, which predicts hazard from future administration and warrants prevention or specific treatment, or alteration of the dosage regimen or withdrawal of the product.” Essentially, an ADR is a subset of adverse drug events (ADE) that represents harm from appropriate drug use, often directly attributed to a drug’s properties (suspected causal association) and *not* related to inappropriate dosing, misuse, or overdose [6].

Classification by type

ADRs can be classified by type, onset, or severity to aid in comparisons among drugs and to guide treatment strategies. As described in Chapter 15, the traditional classification system introduced by Rawlins and Thompson [14] classifies ADRs into Type A and Type B. Type A ADRs are dose related and often predictable as they are a known augmentation of a drug’s pharmacologic effects. Type B ADRs are not dose related and are unpredictable (bizarre) and often rare. Our current understanding of the mechanistic basis of Type B reactions is extensively discussed in Chapter 15. The Type A and B classification system evolved over the years as new drug adverse events became apparent to include Type C through E that incorporate time-related ADRs, withdrawal, and failure of therapy reactions [13, 15]. Type C reactions relate to both dose and time with chronic use and correlate with a cumulative dose (e.g., hypothalamic-pituitary adrenal axis suppression with corticosteroid use) [13]. Type D reactions are time delayed, becoming apparent some time after drug use, such as carcinogenesis [13, 16]. Type E or end-of-use reactions are a consequence of withdrawing a drug such as the withdrawal syndrome that occurs when opiates are suddenly removed [13]. Table 26.1 lists the types of ADRs with additional examples. Although these expanded classification types may not be commonplace, they provide knowledge on possible mechanisms and management [17].

ADR classification systems continue to evolve and another classification system known as **Dose, Time, Susceptibility (DoTS)** similarly classifies reactions based on dose and time and also includes genetic, pathological, and other biological differences to provide a more clinical pharmacological approach [18, 19]. Additionally, a mechanistic classification system known as EIDOS was established to complement the DoTS system and describes mechanisms for the adverse effect: extrinsic chemical species that initiates the effect, intrinsic chemical species that it affects, distribution in the body, the physiologic outcome, and sequela (the adverse effect) [20]. These systems have been used together to assess preventability of an ADR [21].

The traditional classification systems described by Rawlins and Thompson, and others, were formulated based on ADRs induced by small molecule drugs and are not well suited for larger molecules such as biological products (biologics). These biologic therapies include cytokines, antibodies, and fusion proteins. They differ from small molecule drugs because they are larger sized proteins, similar in structure to naturally occurring proteins, administered parenterally, and not metabolized, but rather they are processed [22]. With the increasing use of biologics to treat various diseases, a new classification scheme is warranted to accommodate the different characteristics of these products. Pichler et al. [17] proposed a unique classification of adverse reactions to biologics based on their immune target or mechanism of action. To distinguish it from the traditional classifications, Greek letters are used for the five types as presented in Table 26.2 [22]. This classification system can be used in surveillance of adverse reactions as new biological products are marketed.

TABLE 26.1 Types of adverse drug reactions.

Type	Description	Example
Type A ^a	Extend directly from a drug's pharmacologic action Dose related Predictable Avoidable	Hypoglycemia from insulin Bleeding with anticoagulants
Type B	No relationship to the usual pharmacologic effect of a drug Relatively uncommon Difficult to predict Not dose related, host dependent Includes idiosyncratic reactions, drug allergies	Drug-induced rashes Increased vulnerability of erythrocytes to oxidative injury by several drugs in G6PD-deficient individuals
Type C	Related to the duration of therapy and to some extent, the dose Long-term exposure required	Tardive dyskinesia seen with antipsychotic medications
Type D	Observed with prolonged exposure to a drug or exposure at a critical time	Increased risk of endometrial cancer with tamoxifen
Type E	Observed on abrupt discontinuation of long-term therapy	Delirium tremens on alcohol withdrawal Rebound hypertension with clonidine withdrawal

^aMost common type.

Modified from reference: Rehan HS, Chopra D, Kakkar AK. Physician's guide to pharmacovigilance: terminology and causality assessment. Eur J Intern Med 2009;20(1):3–8 and Pichler W.J. Adverse side-effects to biological agents. Allergy. 2006;61:912–920 (anticoagulation example).

TABLE 26.2 Proposed classification of adverse reactions to biologic agents.

Type	Terminology	Example of reaction and (medication)
α	High cytokine and cytokine release syndrome	Cytokine release syndrome (Muromonab-CD3)
β	Hypersensitivity	Common acute infusion reactions (Rituximab) Delayed reactions (Etanercept) Anaphylaxis (Omalizumab)
γ	Immune or cytokine imbalance: Immunodeficiency	Increased risk of tuberculosis (TNF inhibitors) Hypogammaglobulinemia (Rituximab)
	Autoimmunity	Systemic lupus erythematosus (Interferon- γ)
	Atopic disorders	Atopic dermatitis (TNF inhibitors)
δ	Cross-reactivity	Acne from anti-EGFR (Cetuximab)
ϵ	Nonimmunological side effects	Neuropsychiatric side effects including confusion or depression (Interferon- α)

Abbreviations: EGFR, epidermal growth factor receptor; TNF, tumor necrosis factor.

Modified from reference: Pichler W.J. Adverse side-effects to biological agents. Allergy. 2006;61:912–920.

Classification by severity

Classification of ADRs by severity is helpful in guiding acute treatment and future medication therapy decisions. *Severity* is used to describe the intensity of an ADR that may or may not be serious [13]. For example, a headache may be severe in intensity but may not be life threatening. A severity grading system initially established by the National Cancer Institute for clinical trials with oncology drugs, known as NCI Common Terminology Criteria for Adverse Events (CTCAE) is a descriptive grading system that is used for Adverse Event (AE) reporting [23]. A scale for grading severity is provided for each AE that is listed under the Medical Dictionary for Regulatory Activities (MedDRA) hierarchy System Organ Class (SOC), with each grade referring to the severity of the AE [24]. The grading scale has a range from 1 through 5 with unique clinical descriptions of severity for each AE. Not all AEs are assigned every grade, for example Stevens-Johnson Syndrome can only be assigned grades 3 through 5. Although initially established for comparing AEs for oncology drugs, this severity system can be applied to other drug categories as well. This severity scale can be used to describe any AE regardless of whether or not the AE is causally related to the drug.

Regulatory authorities classify ADEs based on seriousness. FDA defines a *serious adverse drug experience* as one “occurring at any dose that results in any of the following outcomes: Death, a life-threatening adverse drug experience, inpatient hospitalization or prolongation of existing hospitalization, a persistent or significant disability/incapacity, or a congenital anomaly/birth defect” [11]. Important medical events that do not result in death, are not life threatening, or do not require hospitalization may also be considered a serious adverse drug experience when, based upon appropriate medical judgment, they may jeopardize the patient and may require medical or surgical intervention to prevent one of the outcomes listed in the earlier definition. An example of an important medical event is allergic bronchospasm requiring intensive treatment in an emergency department that may not result in inpatient hospitalization. Other regulatory authorities use similar definitions: for example, the European Medicines Agency defines a serious adverse reaction as “an adverse reaction that results in death, is life-threatening, requires hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, or is a birth defect” [25]. International regulatory authorities often use standardized terminology, such as defined by MedDRA, to improve reporting consistency and research regarding ADEs and ADRs [24]. Seriousness of an ADE or ADR is used to prioritize safety issues and determine regulatory action.

The aim of classifying ADRs is to better characterize them for detection, monitoring, and management in clinical trials and subsequent patient care. Some ADRs are predictable and potentially avoidable, such as Type A reactions which comprise about 80% of ADRs [3,26]. Our understanding of a majority of Type B reactions is evolving as described in Chapter 15. Additionally, biological products differ from small molecule drugs and have their own mechanistic ADR classification system (Table 26.2) [17]. General knowledge of ADR mechanisms is helpful in facilitating their detection, treatment, and prevention.

Assessing ADRS

ADR detection

Recognizing ADRs in the clinical setting is challenging because drug-induced illness may closely resemble natural disease. A patient's new symptom, an abnormal laboratory value, or a directly observed event can potentially be an ADR, but it may also be attributable to alternative etiologies such as comorbidity or other treatments. Electronic health records (EHR) and other health technologies provide opportunities to actively detect ADRs [27]. The first step is to conduct a thorough medication history and include ADRs in the differential diagnosis when evaluating a patient with new onset symptoms [20]. In a recent analysis, the prevalence of ADEs was approximately 6% in patients hospitalized for a cardiovascular event, pneumonia, or conditions requiring surgery [28]. Elucidating additional details related to the symptoms before instituting treatment with an additional medication may uncover an ADR that is responsible for the symptoms and prevent adding unnecessary medications or prolonging the ADR.

Some methods of retrospectively detecting ADRs in the inpatient and outpatient setting include medication order screening for specific medication changes or alert drugs. Abrupt discontinuation of a drug and an order for a corticosteroid or diphenhydramine may indicate a drug allergy. A dose reduction or an order to “hold” a drug may prompt a pharmacist to check for an abnormal drug level and screen for a potential ADR or drug interaction. Trigger tools have been developed to detect ADRs, to help characterize the most frequent AEs and to develop priorities for quality improvement and prevention [29,30]. These methods involve a two-stage process: a screening process for the detection of criteria or triggers, and subsequent in-depth manual review of the medical record for an ADE [30]. Additionally, newly approved medications can be targeted for medication order screening for safety concerns identified during premarketing trials and noted in drug labeling

that may not have been fully characterized. Other strategies for ADR detection and reporting include targeting medication classes most commonly associated with ADEs such as antibiotics, corticosteroids, antineoplastics, anticoagulants, NSAIDs, and narcotics [31,32]. Tools have also been developed to identify patients at higher risk for certain ADRs based on EHR information. These tools incorporate patient demographics, medications, medical history, and laboratory results into predictive models that may facilitate ADE prevention and mitigation in real time [33–35].

An abnormal laboratory value or diagnostic test can alert a health care provider to review the patient's EHR for an ADR. EHRs can be programmed to compile reports of laboratory values that are outside a predetermined range. For example, alanine aminotransferase (ALT) elevation of three times the reference range can alert a health care provider to review for any drug-related causes such as a statin. Baseline laboratory testing and subsequent laboratory monitoring is used to monitor for ADRs with certain drugs. Changes in laboratory values do not necessarily mean that an ADR has occurred but can alert a health care provider to review a patient's medication record and ask the patient pertinent questions to assess for an ADR.

Another way of identifying ADRs is through review of daily inpatient multidisciplinary notes or screening of outpatient clinic notes. Reviewing notes for lethargy, oversedation, and cognitive decline may be a sign of an ADR caused by a sedative, analgesic, or hypoglycemic agent. Multidisciplinary notes may provide additional information on how an event was managed and if an ADR was considered or if an alternative medication was added to treat the event, which could present opportunity for additional ADRs. This process is time consuming and may be an opportunity for new technologies to enhance detection using natural language processing [36].

Special populations such as infants, children, pregnant women, and the elderly are susceptible to ADRs. These populations are often not well studied during premarket clinical trials and changes in development, reduced drug clearance secondary to impaired renal or hepatic function, and new onset comorbidities can increase the risk of an ADR [19]. Mitigation strategies can be implemented to circumvent ADRs such as routine screening of prescription orders for appropriate dose when administering medications requiring renal dose adjustments (e.g., gabapentin, ganciclovir, ranitidine, and rivaroxaban) to a patient with changing renal function. Computerized physician order entry (CPOE) systems with decision support that incorporates laboratory data can alert health care providers to adjust the dose of renally and hepatically eliminated drugs during the ordering process thus preventing an ADR. Knowing a drug's dose-related ADRs and monitoring for them can avoid harm.

Assessing risk and causality

Determining if a patient's AE is an ADR is complex and requires a causality assessment. Patients often take more than one drug and have several comorbidities making causality assessment challenging. Causal association assessment in the clinical setting is focused on a particular individual who experienced the AE and often it is not possible to know with a high level of certainty whether the AE was caused by the drug. Clinical data from the patient, pertinent laboratory results, and extensive drug information review aid in determining whether there is a causal association between a drug and an AE. Ultimately the determination is based on the strength of evidence from all available data. The Bradford Hill criteria for causation, originally presented to demonstrate epidemiologic evidence of a causal relationship between an exposure and outcome, is useful in deciding if some level of causal association is present [37]. Table 26.3 is a causality checklist with some of the principles from the Bradford Hill criteria, which include temporality (the event occurs after drug initiation), positive dechallenge (the event resolves with drug discontinuation), positive rechallenge (event reappears with drug reinitiation), and ruling out other plausible causes. Edwards and Aronson [13] outlined steps to assess causality that include (1) thorough medication history (including over-the-counter medications), (2) timing of medications to the event (latency), (3) pattern of the AE to the known pharmacology or allergy pattern of suspected drugs, (4) background frequency of the event, (5) investigations (drug levels, organ function tests, biopsies) to aid in diagnosis or rule out alternative causes, and (6) consideration of rechallenge if appropriate and if patient will directly benefit from knowledge gained. When assessing a potentially drug-related AE or symptom, review for latency of development of the AE after drug initiation in the clinical context of the event, and the known pharmacokinetics (PK) and pharmacodynamics (PD) of the drug, for plausibility. For example, the onset of the reaction should be clinically reasonable such as an anaphylactic reaction, which usually occurs within a few minutes to several hours after the drug is administered versus agranulocytosis, which may take days to weeks to develop. If a drug is suspected of causing cancer, which may take a much longer time to develop, a latency of less than a month would exclude causality. Additionally, it is necessary to review recognized adverse reactions of the drug, or related drugs in the same class, and if the AE under assessment is similar or involves similar mechanisms, because a causal association would be more likely.

TABLE 26.3 Causality considerations.

Patient-level data (intrinsic evidence considerations)
Temporality
Dechallenge/rechallenge
Response to antidote/reversal agents
Alternative causes
Drug levels in body fluids or tissues
Dose-response relationship
Prior adverse event experience with the same or a similar drug
Existing knowledge about the potential ADR (extrinsic evidence considerations)
Known association with drug or drug class
Strength of association
Disease/event epidemiology
Biologic and pharmacologic plausibility

A thorough review of available drug information is needed to assess the possible role of a drug in causing an AE. It is logical to start with a review of the safety information in the Boxed Warning, Warnings and Precautions, and Adverse Reaction sections of the drug's labeling (also known as the package insert or the prescribing information), which can be found at Drugs@FDA or DailyMed [38, 39]. Drugs@FDA, which contains the most recent FDA-approved drug labeling, also contains redacted premarketing review documents from FDA medical, chemistry, and pharmacology reviewers as well as other disciplines that may contain information to establish biologic plausibility for a drug-AE association. DailyMed, sponsored by the National Library of Medicine, contains drug labeling but may also contain any labeling changes that have been recently submitted to the FDA by the drug manufacturer^a that are under review. MedWatch safety alerts contain information from reports submitted to the FDA Adverse Event Reporting System (FAERS) from health professionals, patients, consumers, and drug manufacturers [40]. Additionally, FDA posts potential signals of serious risks/new safety information identified from FAERS in quarterly reports [41]. A review of published medical literature for similar case reports can provide additional evidence to support an association.

Causality assessment is needed to classify the relationship between a drug and AE at two levels: the individual patient case and the overall drug-AE level. Assessment of a causal association at the drug-AE level considers the strength of evidence collectively from various sources of data and is discussed later in this chapter. Causality assessment at the individual patient level uses clinical judgment based on assessment of the AE in the context of what is known about the patient and the drug [42]. Methods have been developed to standardize assessment in the clinical setting, which use different causality categories and are assessed using different criteria. Some methods are specific to a particular ADR while other methods are more general, and some can be complex and time consuming [43]. Two approaches that are easy to use are WHO-Uppsala Monitoring Centre (WHO-UMC) Causality Categories (Table 26.4) and the Naranjo probability scale (Table 26.5). The WHO-UMC Causality Assessment relies on expert judgment in considering the clinical and pharmacological aspects of the case and the quality of the observation [43]. This system assigns one of six category terms to the drug–event combination based on several assessment criteria: certain, probable/likely, possible, unlikely, conditional/unclassified, and unassessable/unclassifiable [44]. AEs with a probable/likely or certain causal association are likely to recur, therefore knowing an event is probable/likely, may help prevent future episodes [45]. In practice, few ADRs are classified as “certain” because information relevant to assessing all alternative etiologies may not be available and it may not be practical to obtain confirmatory evidence (e.g., biopsy); therefore, most drug-event combinations with a causal association deemed greater than unlikely are considered possible or probable.

a. Throughout this chapter the term “drug manufacturer” is used for simplicity, but is intended to encompass sponsors, applicants, licensed manufacturers, packers, distributors, and other responsible persons subject to FDA's requirements for postmarketing safety reporting.

TABLE 26.4 WHO-UMC causality categories.

Causality term	Assessment criteria ^a
Certain	• Event or laboratory test abnormality, with plausible time relationship to drug intake • Cannot be explained by disease or other drugs • Response to withdrawal plausible (pharmacologically, pathologically) • Event definitive pharmacologically or phenomenologically (i.e., an objective and specific medical disorder or a recognized pharmacological phenomenon) • Rechallenge satisfactory, if necessary
Probable/likely	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Unlikely to be attributed to disease or other drugs • Response to withdrawal clinically reasonable • Rechallenge not required
Possible	• Event or laboratory test abnormality, with reasonable time relationship to drug intake • Could also be explained by disease or other drugs • Information on drug withdrawal may be lacking or unclear
Unlikely	• Event or laboratory test abnormality, with a time to drug intake that makes a relationship improbable (but not impossible) • Disease or other drugs provide plausible explanations
Conditional/ unclassified	• Event or laboratory test abnormality • More data for proper assessment needed, or • Additional data under examination
Unassessable/ unclassifiable	• Report suggesting an adverse reaction • Cannot be judged because information is insufficient or contradictory • Data cannot be supplemented or verified

^aAll points should be reasonable complied with.

Reproduced from World Health Organization. The use of the WHO-UMC system for standardised case causality assessment. <https://www.who.int/publications/m/item/WHO-causality-assessment>. WHO 5 June 2013 [Accessed 30 June 2020].

The Naranjo probability scale is a quantitative instrument for causality assessment that is commonly used in the hospital setting and utilizes similar principles as the WHO-UMC system. By answering 10 questions regarding onset of the reaction, investigations, dechallenge and rechallenge information, alternative causes, and prior exposure, an ADR numeric probability score is assigned and classified based on the total score (Table 26.5) [46]. Unfortunately, no scale can provide a definitive causality ascertainment and some tools may not work well for certain ADRs such as Stevens-Johnson Syndrome (SJS) and toxic epidermal necrolysis (TEN) [47]. However, they can guide health care providers through the thought process for assessing ADRs. Determining if the patient is experiencing an ADR by using causality assessment tools can prevent patient exposure to additional often unnecessary medications, determine appropriate treatment, ensure proper documentation of the ADR diagnosis, and lead to measures to prevent future occurrences (e.g., use a lower dose or pretreat with an antihistamine) [6, 48].

Managing and reporting ADRS

Managing ADRs

Despite efforts to avoid or minimize ADRs, they still occur and may require swift action to avoid harm. For serious, life-threatening ADRs such as toxic epidermal necrolysis, the offending drug should be stopped and supportive care provided [49]. ADR management can be guided by the type of ADR, for example, Type A reactions are dose dependent and may respond to a dose reduction whereas Type C reactions that occur with long-term use can be managed by using the lowest effective dose or the shortest duration of therapy for the clinical condition, such as with corticosteroids and adrenal axis suppression. Some ADRs may require specific reversal agents such as idarucizumab for the treatment of uncontrolled bleeding with dabigatran [19]. Information on how to treat or otherwise manage an ADR that has occurred may be found in a drug's prescribing information [50, 51]. While information may be acquired from multiple sources, answers are often

TABLE 26.5 The Naranjo Adverse Drug Reaction Probability Scale.

To assess the adverse drug reaction, please answer the following questionnaire and give the pertinent score	Yes	No	Do not know	Score
1. Are there previous <i>conclusive</i> reports on this reaction?	+1	0	0	
2. Did the adverse event occur after the suspected drug was administered?	+2	−1	0	
3. Did the adverse reaction improve when the drug was discontinued or a <i>specific</i> antagonist was administered?	+1	0	0	
4. Did the adverse reaction reappear when the drug was readministered?	+2	−1	0	
5. Are there alternative causes (other than the drug) that could have on their own caused the reaction?	−1	+2	0	
6. Did the reaction reappear when a placebo was given?	−1	+1	0	
7. Was the medication detected in the blood (or other fluids) in concentrations known to be toxic?	+1	0	0	
8. Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	+1	0	0	
9. Did the patient have a similar reaction to the same or similar drugs in <i>any</i> previous exposure?	+1	0	0	
10. Was the adverse event confirmed by any objective evidence?	+1	0	0	
The ADR is assigned to a probability category from the total score as follows: <i>definite</i> if the overall score is 9 or greater, <i>probable</i> for a score of 5–8, <i>possible</i> for 1–4 and <i>doubtful</i> if the score is 0.			Total	

Reproduced with permission from Naranjo CA, Busto U, Sellers EM, Sandor P, Ruiz I, Roberts EA, Janecek E, Domecq C, and Greenblatt DJ. A method for estimating the probability of adverse drug reactions. Clin Pharmacol Ther 1981;30:239–245.

lacking to many practical questions regarding management, such as if continued use will lead to progression or regression of an ADR, or how long after drug discontinuation will it take for an ADR to resolve. Insight to answer some of these questions may be provided by a drug's clinical pharmacology and a search of the medical literature for case reports of similar drug-events for possible management options. Whenever it is reasonably clear that a specific drug is causal, there should be a reevaluation for its need and consideration of therapeutic alternatives with an unlikely risk of the ADR [13]. Although one might assume an ADR to a drug might translate to avoidance of all drugs in the same class, this is dependent on the type of reaction, underlying mechanism of the reaction, and availability of alternative therapies. For example, drug allergy with a penicillin does not automatically mean the patient is allergic to all beta-lactam antibiotics as new generations of beta-lactam containing molecules with dissimilar side chains, such as cephalosporins exhibit less (or no) cross-reactivity [52]. If certain drug hypersensitivity reactions occur with a drug that is needed and for which there is no alternative, desensitization can be attempted under supervision of an allergist/immunologist [53]. When a patient is taking multiple drugs that could cause the ADR, eliminating those drugs (dechallenge) that are least essential to care is a reasonable approach.

Reporting ADRs

It is necessary to document and report ADRs in order to evaluate the benefit-risk profile of a drug and develop mitigation strategies. Various accrediting organizations, societies, and regulatory authorities, both in the United States and internationally, have developed guidelines for reporting ADRs [54–56]. As it may be impractical to report every ADR that is encountered in clinical practice, some authors have encouraged reporting just those ADRs that are serious, associated with a drug that was recently marketed (1–3 years), and considered high risk by regulatory and medical institution accreditation organizations, such as anticoagulants, insulins, and narcotics [57]. Nonetheless, spontaneous reporting of ADRs by health care providers, patients, and caretakers continues to be an important source of drug safety signals. Publishing case reports of ADRs also disseminates information to other health care providers, regulatory authorities, and stakeholders and provides for detailed information about the AE and drug association.

The FDA supports direct reporting of observed or suspected adverse events to the Agency via the MedWatch program. FDA launched the MedWatch program in 1993 as part of a medical product safety reporting program [58]. The program is still active today with over 21 million reports of adverse events, medication errors, and drug quality complaints captured in the FAERS database as of 2020. The MedWatch program enables submission of voluntary reports via phone (toll-free: 1-800-FDA (332)-1088), fax (1-800-FDA(332)-0178), or online via the MedWatch website <https://www.fda.gov/safety/medwatch-fda-safety-information-and-adverse-event-reporting-program> [59]. The original goals of the MedWatch program were to make it easier for health care providers to report serious events for any FDA-regulated medical product, clarify which types of reports the FDA wants to receive, disseminate information on the FDA's actions that have resulted from AE reporting, and increase understanding and awareness of drug and device-induced disease [58]. In the years since its inception, MedWatch has broadened its reach with expanded tools and processes for disseminating safety information to both health care providers and consumers [60].

Because spontaneous reporting systems depend upon voluntary participation, FAERS is limited by underreporting, selection bias, and variability in report quality. Factors influencing AE reporting tend to relate to motivations and attitudes. Inman [61] proposed the “seven deadly sins” theoretical basis for the underreporting of AEs among health care providers (financial incentives, legal aspects, complacency, diffidence, indifference, ignorance, and lethargy). A systematic review of the determinants of underreporting of adverse reactions by Lopez-Gonzalez et al. [62] confirmed that Inman's model can account for factors influencing ADR underreporting. Barriers to reporting include lack of awareness of the reporting system, inadequate skills needed to report, erroneous beliefs about consequences, motivation and goals, social influences, and environmental constraints [63]. Moreover, spontaneous reports may have significant variability in the quality of information in the report as some reports have detailed information while others may be missing the critical information needed to assess a causal relationship. Nonetheless, spontaneous reporting of ADRs has enabled the detection of drug-related risks.

Pharmacovigilance: Postmarketing surveillance of adverse drug reactions

Pharmacovigilance, as defined by the WHO, is the science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other possible drug-related problem [8]. Pharmacovigilance systems are designed to detect ADRs that were not noted during a drug's preapproval clinical development program and to identify medication errors and problems with drug quality [64]. In the United States, the FDA is responsible for protecting the public health by ensuring the safety, efficacy, and security of human and veterinary drugs, biological products, and medical devices. As part of this charge, FDA conducts surveillance of data for new safety information or potential signals of a serious risk [65]. Other countries have developed ADR reporting systems to assist and identify ADRs which include: Canada Vigilance Program; United Kingdom Yellow Card Scheme operated by the Medicines and Healthcare Products Regulatory Agency (MHRA); Australia's Blue Card system; the French Pharmacovigilance Spontaneous Reporting System database; the Adverse Drug Reaction Information Management System of the Pharmaceutical and Medication Devices Agency; Ministry of Health, Labor, and Welfare (Japan); the Lareb database (Netherlands); and the BiSi database (Sweden). These ADR reporting systems as well as other pharmacovigilance activities have expanded over the years in response to tragic events resulting from drug toxicity [66]. The FAERS will be used as an example throughout the remainder of this chapter.

A safety signal is information that arises from one or multiple sources (including observations or experiments), which suggests a new, potentially causal association between a drug and an AE. Signals can also be a new aspect of a known association between an intervention and an event or set of related events, either adverse or beneficial [67]. Signals can arise from various postmarket data sources (e.g., case reports, observational studies), preclinical data, and events associated with other drugs in the same pharmacologic class [64]. Identifying safety signals in these data sources is known as signal detection. Even a single well-documented case report can provide a safety signal, particularly if the report describes a positive rechallenge (i.e., following drug discontinuation and recovery from an event, the drug is restarted and the event reoccurs) or if the event is extremely rare in the absence of drug use [68]. Once a safety signal is identified, further evaluation of a theoretical association is warranted.

Experience drawn from the postmarketing setting can lead to regulatory actions such as revising the drug's labeling to include new information on drug risks and benefits to health care providers and patients. This information may affect a health care provider's decision to prescribe a drug or a patient's willingness to take the drug. A review of drugs approved by the FDA between 2002 and 2014 determined labeling updates occur frequently following a drug's approval with the period between the second and eighth postapproval years being the most active [9]. Importantly, labeling updates for drugs also occur outside of this time period, signifying that surveillance must continue throughout the drug's life cycle [69].

Postmarketing surveillance principles from an FDA perspective

FDA monitors the benefits and risks of drugs throughout their life cycle and takes regulatory action(s) when there is a new or increased risk. FDA’s safety surveillance starts early in the drug’s life cycle as part of the drug application review process that may lead up to FDA approval. A multidisciplinary team evaluates the drug applications filed with the FDA, and considers appropriate measures to assess the safety of the drug in the postmarketing period. Multidisciplinary team members bring their expertise in medicine, pharmacology, epidemiology, safety surveillance, medication error prevention, risk management, product quality, and statistical analysis together to develop a postmarketing surveillance strategy that is drug specific using a risk-based approach [70,71].

The principles of risk-based safety surveillance consider the drug’s characteristics and use and incorporate this information in making decisions on the appropriate frequency and extent of monitoring. Drugs that generally are subject to more extensive monitoring are included in Table 26.6. When conducting surveillance, safety specialists prioritize their review of information that suggests a safety signal or broadly describes safety concerns (i.e., important identified risk(s), important potential risk(s), and important missing information) for the drug being evaluated [72]. Safety information of interest to safety specialists during surveillance is listed in Table 26.7 [71].

After identifying a suspected new risk with a drug, the FDA evaluates the potential risk using available evidence from various data sources. Evaluations may include analyses of drug utilization data, the FDA’s Sentinel system (see

TABLE 26.6 Drug characteristics that may require extensive postmarketing surveillance [71].

New molecular entities and original biological license applications
Biosimilar biological products
First in class approvals
Newly approved formulation(s)
Newly approved indication(s)
Extension into new patient populations
Drugs with complex pharmacokinetic (PK) or pharmacodynamic (PD) characteristics
Drugs with complex compositions or manufacturing processes

TABLE 26.7 Safety information aspects used in postmarket surveillance of drugs [71].

Important potential risks of the drug recognized at the time of or after approval
Apparent increase in the severity or frequency of reporting of a labeled AE
Deaths, particularly in populations or in patients using the drug for indications for which there would not be the expectation of death
AEs for which causal attribution to the drug is biologically plausible, based on the product’s known pharmacological action
Reports of unlabeled, serious AEs
Serious AEs thought to be rare in the general population and associated with a high drug-attributable risk
Interactions among different drugs (e.g., drug-drug, drug-device, drug-food, or drug-dietary supplement)
Reports of reduced effectiveness or efficacy
Medication errors resulting from confusion about a drug’s name, labeling, packaging, use
Off-label use, misuse, abuse, and other intentional uses of a drug in a manner that is inconsistent with the FDA-approved labeling
AEs reported or observed in a specific patient population
AEs for which a Risk Evaluation and Mitigation Strategy (REMS) is intended to mitigate the risk
AE, adverse event.

“Epidemiologic assessments” later in this chapter), clinical pharmacology data, mechanistic studies, or other population-based approaches to support or refute the risk [71]. The FDA can also ask drug manufacturers to perform additional studies if the information is unknown and a study is feasible [73]. The next section discusses additional data sources.

Data sources for signal identification

FAERS

For several decades, spontaneous reporting systems such as FAERS and its predecessor systems have been the main sources of ADR information. An analysis of all US drug labeling changes occurring in 2010 found that 52% of labeling changes were informed by spontaneous reports [74]. Similarly, an evaluation of all FDA-issued Drug Safety Communications between 2007 and 2009 found that 57% were informed by FAERS data [75]. FAERS relies on health care providers, consumers, patients, and others to report (AEs) voluntarily either to the drug’s manufacturer, which will subsequently report them to the FDA according to regulations, or to the FDA directly via the MedWatch program. FAERS is a computerized database containing over 21 million AE individual case safety reports (ICSRs) received from 1969 through 2020 [76]. ICSRs in FAERS contain narrative text describing an AE, a list of suspected drugs involved in that event, and information identifying the reporter. Reporters may elect to supply additional information such as past medical history, laboratory data, details about the suspect drug, concomitant medications, event dates, exposure dates, and the event outcome (e.g., hospitalization, death). The structure of FAERS complies with international safety reporting guidance by the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) [77].

Data mining

Multiple similar case reports or even a single well-documented case report can indicate a signal, especially if the report describes a positive rechallenge or if the event is extremely rare in the absence of drug use. Because of the increasingly large volume of ICSRs received annually (e.g., >2 million in 2020), FDA screens ICSRs and uses disproportionality analyses as a data mining tool to identify potential signals. Data mining refers to the use of statistical or mathematical tools to discover patterns of associations or unexpected occurrences in large databases [78]. It involves the systematic examination of reported AEs to estimate if the AE is reported at a higher frequency for a particular drug relative to other drugs in the database (disproportionality). Unexpectedly high reporting associations may generate a hypothesis that there may be an association between the particular AE and the drug, but it is not a confirmation of a causal association. Similarly, the absence of disproportionality does not confirm the absence of a safety signal or disprove a signal detected by other methods.

Limitations of spontaneous reporting systems

While FAERS is an important source of postmarketing safety information, it does have limitations. Because of the voluntary nature of reporting, the magnitude or incidence of risks cannot be determined from FAERS data. Various factors such as the duration of drug marketing, severity of the AE, reporting regulations, and publicity influence whether an AE will be reported [79]. Comparisons between drugs can be challenging particularly if their marketing periods differ and if certain factors influencing reporting are present. For example, drug manufacturer-sponsored programs, such as patient support programs, have increasingly contributed to the number of ICSRs in the FAERS database [80]. FDA issued a final rule in 2014 requiring drug manufacturers to submit postmarketing safety reports in an electronic format that the Agency can process, review, and archive [81]. Implementation of the electronic reporting rule resulted in the electronic submission of ICSRs that may have previously been submitted on paper [82]. The quality of ICSRs in terms of the completeness of the report and relevance of information provided varies among reports. An analysis of all ICSRs reporting an outcome of death received in FAERS as of 2017 demonstrated that key information needed to assess the role of a drug in a patient’s death was frequently missing (e.g., patient’s past medical history and concomitant medications) [80]. Bohn et al. [83] demonstrated that it was difficult to accurately identify a drug’s manufacturer when multiple generic drugs with the same active ingredient are marketed, therefore making it difficult to evaluate potential drug quality issues with FAERS data. So, techniques are being developed to identify ICSRs with more robust information and differentiate them from less informative cases to improve timeliness in assessing safety signals [84]. Lastly, ICSRs describing the same occurrence of an AE in a patient reported by more than one reporter can result in duplicate reports in FAERS. The reporter could be drug manufacturers of the different drugs when there are multiple suspect drugs, different drug manufacturers of a single suspect drug (e.g., brand and generic drug manufacturers), or reports directly from the public to the FDA that were also reported by a drug manufacturer. During case series evaluation, it is not unusual for 10%–25% of ICSRs for a case series to be duplicate reports [84]. These limitations need to be considered in the evaluation of potential safety signals.

Other data sources for signal identification

There are many other information sources for safety signal identification which can include medical literature monitoring, review of a drug manufacturer's Periodic Safety Reports (PSR), postmarketing studies, and communications with international regulators. Routine screening of the medical literature is helpful to identify emerging safety signals that are not submitted as ICSRs to FDA or other regulatory authorities. Screening is accomplished by searching the medical literature by the drug name, drug class, or by an AE of interest. The principles underlying how safety specialists select drugs and AEs for medical literature screening are listed in [Tables 26.6 and 26.7](#) [71]. Safety specialists supplement their screening of published AE case reports with additional data sources, such as studies completed by academic institutions or other researchers outside of FDA, studies performed by other Government agencies, and results from randomized controlled clinical trials that were required of the drug manufacturer or the drug manufacturer voluntarily performed in the postmarket setting. Information on the use of the drug in special populations (e.g., pediatrics, geriatrics, patients with chronic kidney disease) may be available in postmarketing studies. These may be presented in peer-reviewed or online journals or as abstracts at conferences. Safety specialists also communicate with international regulators on any trends in AE reporting especially if the drug was approved earlier in another country. Additionally, drug manufacturers submit PSRs, which may include safety assessments of previously unknown risks that they are monitoring. The drug manufacturer is required to submit PSRs to FDA on a recurring basis for review [85]. These PSRs provide summary information directly from the drug manufacturer, which may include clinical and nonclinical study reports and the drug manufacturer's assessment of the marketed drug's benefit-risk profile. In reviewing PSRs, reviewers pay particular attention to drugs with recently approved new indications or worldwide safety data with findings that might suggest an emerging safety signal.

Signal evaluation

Safety signal identification merely establishes a hypothesis that an ADR exists and requires subsequent evaluation and understanding of the safety risk. Identified safety signals are prioritized based on the AE, the seriousness of the outcome, how long the drug has been marketed, and the public health impact. Evaluations are conducted to see if the benefit-risk profile of a drug has changed based on the new information. A multidisciplinary team evaluates the prioritized safety signal using various data sources to determine whether regulatory action(s) are indicated. The data sources that are collected and analyzed can include, but are not limited to, ICSRs submitted to FAERS, medical literature case reports, observational studies, drug utilization data, and epidemiologic assessments.

Case reports often play a critical role in evaluating a safety signal [74]. Case reports with the AE of interest are retrieved from FAERS and the medical literature to formulate a case series. The search for case reports requires pharmacovigilance expertise and includes consideration of the safety signal to be evaluated, whether or not to include all drugs in the class, and a specified time period in order to retrieve the most pertinent data. Because the case reports that form a case series may derive from different sources (FAERS or medical literature) and from various reporters with varying medical expertise, it is usually necessary to develop a *case definition*. The case definition includes clinical characteristics of the event of interest, without taking into account the causal role of the drug with the AE being investigated [64]. A case definition is developed using information from medical literature and clinical practice guidelines and consists of criteria for determining whether a person should be identified as having a particular disease, injury, or other health condition [70]. Case definitions provide a consistent and reproducible approach to identifying cases of interest. Each report is then reviewed for inclusion in the case series if the case definition criteria are met. As medical literature reports may also be submitted to FAERS and multiple reporters can submit reports for the same patient, these duplicate reports need to be identified and removed. Assessment of causal association is the final step in developing a case series. Specific features that are evaluated include (1) chronologic data (e.g., plausible temporal sequence, dechallenge, rechallenge), (2) precedents (e.g., similar AEs with the same drug or related drug), (3) biological or pharmacological plausibility (e.g., toxic drug concentration in body fluid, occurrence of a recognized pharmacodynamic phenomenon), (4) information quality, and (5) alternative etiologies (e.g., concurrent diseases or conditions, concomitant medications) [71]. Assessing for potential causal association can be challenging; however, it is useful in excluding cases with no temporal relationship such as if the adverse event occurred prior to initiation of a drug. The causal assessment tools presented in [Tables 26.4 and 26.5](#) can be used to strongly suggest an association; however, rarely can information from spontaneous reporting systems provide definitive answers [86]. In addition, summaries for select ICSRs that contain a majority of the elements of a good case report (e.g., time to onset, comorbid conditions, concomitant medications, documentation of diagnosis, response to dechallenge and rechallenge) and best represent the cases in the series are often discussed in detail. The evaluation of the developed case series may reveal important trends, risk factors, vulnerable subpopulations, or other attributes that may provide supportive evidence for the safety signal.

Drug utilization

New ADRs may emerge with increased use of a drug, when used in patients with more comorbidities, or when a drug is used off-label for other conditions or in other patient populations (e.g., in a pediatric patient). Consequently, it is important to know utilization patterns and how the patient and prescriber characteristics may differ from those included in the pre-approval clinical trials. Information from drug utilization analyses aid in identification and evaluation of safety signals and are useful in designing ways to mitigate identified risks. For example, a utilization analysis may identify specific prescribers that may benefit from enhanced communication (e.g., Dear Health Care Provider Letter) about a particular risk.

National utilization data is available in countries with a single-payer system; however, this is not available in the United States and use of data from other countries may not be generalizable to the US population. Sources of drug utilization data include drug manufacturer sales, outpatient prescription transactions, unique patient exposures, office-based physician surveys, hospital discharge billing, and longitudinal health care claims. Depending on the data sources used, the multidisciplinary team can obtain information about patient demographics, prescriber specialty, diagnoses and procedures associated with the patient visit, directions for use, prescribed dosing, dosage form or route of administration, duration of drug use, drugs taken concurrently, and use during pregnancy. Drug utilization data can be used to provide an estimate of the number of prescriptions dispensed and help put FAERS reports into context (e.g., low number of prescriptions dispensed but there are many reports of Stevens-Johnson syndrome or other very rare event) [64]. Each utilization data source has limitations that can impact appropriate interpretation and application of the data. Overall, drug utilization analyses can provide context for other pharmacovigilance activities and provide insight into real-world use.

During evaluation of a safety signal from a postmarketing spontaneous reporting system (e.g., FAERS), a *reporting ratio* (sometimes called a reporting rate) can be estimated by using drug utilization data as a surrogate measure of drug exposure. Reporting ratios are not incidence rates, but can be used to provide context and generate hypotheses using real-world data. Although reporting ratios are useful in supporting safety signal evaluation, they have limitations. The numerator (representing the number of cases of an AE) and the denominator (obtained from the approximate use of a drug in a population) are from different data sources; in addition, there may be underreporting of the AE, and the size of the exposed population is in itself an estimate [70, 71]. For these reasons, reporting ratios are not considered in isolation; safety specialists must take into account all available data and the strength of such evidence. Nonetheless, reporting ratios can be useful crude estimates for providing context and are exploratory theories to the extent that inherent limitations from each data source are considered appropriately (e.g., both numerator and denominator are aligned by date, time period, indication for use, setting of care, and reporting requirement considerations) [71].

Epidemiologic assessments

Other data sources, including well-designed pharmacoepidemiology or clinical studies, may be necessary to assess safety signals if information from a case series is insufficient or additional information is needed to strengthen the evidence. Epidemiologic assessments are often an integral part of the signal evaluation process. In the past, epidemiology studies were conducted in single health care systems to evaluate a given safety issue with a drug; however, with the passing of the FDA Amendments Act (FDAAA) in 2007, FDA was required to develop methods to gain access to a variety of data sources and validate methods for the establishment of a system to link and analyze safety data from multiple sources [87]. This is known as the Sentinel Initiative and the sources include data from public, academic, and private entities. Through the Sentinel Initiative, FDA accesses information from large electronic health care databases, such as electronic health records, insurance claims data, and registries [88, 89]. These health care databases are made available through a distributed data system that enables FDA to actively monitor the safety of marketed drugs. The Sentinel Initiative began as a pilot (Mini-Sentinel) and has since evolved into a fully integrated system that allows for both active surveillance and methodological innovation and serves as a platform to advance the science of real-world evidence. Hundreds of safety analyses have been conducted by FDA since the official launch of Sentinel's Active Risk Identification and Analysis (ARIA) system in 2016 [90]. While FDA considers the totality of available information when taking a regulatory decision such as drug withdrawal, analyses from ARIA have provided significant contributions.

Regulatory actions

Based on the totality of evidence evaluated from the various data sources mentioned before, the FDA may take regulatory action to reduce the potential for harm or communicate new risk information to health care providers and the public. The multidisciplinary team may determine certain actions are necessary such as (1) requiring a change to the drug labeling, (2) issuing a drug safety communication, (3) gathering additional data through requiring a postmarketing study or trial with

the aim of better characterizing the risk, and (4) requiring a new or modifying an existing Risk Evaluation and Mitigation Strategy (REMS) to better mitigate the risk.

If insufficient evidence exists to support a causal association between the drug and an AE, the AE can be considered an AE of interest for continued close monitoring. Regardless of regulatory action taken, FDA continues to monitor for new safety information. Other regulatory authorities have their own risk communication and mitigation strategies [91]. Some of the strategies FDA uses are briefly described as follows.

Drug labeling changes

FDA-approved drug labeling for health care providers summarizes in an easy-to-read format the essential scientific information that is needed for the safe and effective use of a drug. All new human prescription drugs approved since June 2001 and certain new human prescription drugs approved before June 2001 (e.g., those approved for new uses after June 2001) must have prescribing information in the Physician Labeling Rule format. This effort standardizes the content and format of labeling to make information in the labeling (including safety related information) easier for health care providers to access, read, and use when making prescribing decisions [92]. When new safety information about a drug is identified, FDA can require drug manufacturers to make safety-related labeling changes [93]. FDA guidance documents describe how new safety information is implemented including the location of the information in labeling (e.g., Adverse Reactions, Warnings and Precautions, Boxed Warning sections) [94–96].

Safety communications

Important drug safety issues, including emerging drug safety information, is communicated to health care providers, patients, consumers, and other interested persons in a timely fashion so that they have the most current information about potential risks and benefits of a marketed drug [97]. This information is disseminated as a Drug Safety Communication (DSC), a specific tool used by FDA to communicate important new and emerging safety information. The FDA website contains the most recent Drug Safety Communications as well as links for Early Communications, Follow-Up Early Communications, and Information for Healthcare Professional sheets (<https://www.fda.gov/drugs/drug-safety-and-availability/drug-safety-communications>).

FDA considers issuing a DSC when the information can help with prescribing decisions, affect a patient's decision to use the drug, or indicate whether actions can be taken to avoid, prevent, or minimize harm. For example, eszopiclone, zaleplon, and zolpidem, prescription drugs used for insomnia, received a Boxed Warning for serious injuries caused by sleepwalking, which was communicated as a DSC [98]. DSCs may relate to previously unknown safety concerns including drug interactions, a potential medication error, or updated information about a known AE and highlight potentially serious or life-threatening risks. DSCs include information regarding:

- The safety issue and the nature of the risk being communicated.
- The approved indication or use of the drug.
- The established benefit(s) and risks of the drug being discussed.
- Recommended actions for health care professionals and patients, when appropriate.
- A summary of the data reviewed or being reviewed by FDA.

In addition to DSCs issued by the FDA, the drug manufacturers may be requested or required by FDA to issue a Dear Health Care Provider letter to disseminate information regarding a significant health risk, to highlight important changes in prescribing information, or to communicate corrections to prescription drug advertising or prescribing information.

Postmarketing studies and trials

When new safety information becomes available, the FDA Amendments Act (FDAAA) authorizes FDA to require that drug manufacturers of approved drugs conduct additional studies or clinical trials and establish timelines for their completion [99]. Before requiring a postmarketing study, FDA must find that AE reporting under section 505(k)(1) of the FD&C Act and the postmarket ARIA system, a component of FDA's Sentinel System, will not by themselves be sufficient to meet the purposes of assessing a safety signal [100]. In this situation, FDA can require that an observational pharmacoepidemiologic study or a randomized clinical trial be conducted. For example, upon approval of remdesivir the drug manufacturer was required to conduct a clinical trial to evaluate the pharmacokinetics and safety of remdesivir in subjects with mild, moderate, and severe renal impairment to establish appropriate dosage recommendations in patients with COVID-19 with impaired renal function [101].

Additional pharmacovigilance activities

In an effort to enhance FDA's ability to perform safety surveillance of any AEs of interest, FDA may request that the drug manufacturer:

- Use a targeted data collection tool to gather detailed case information specific to the drug and AE of interest (e.g., liver injury specific questionnaire).
- Submit reports of labeled AEs of interest more frequently and beyond minimum reporting requirements.
- Summarize and assess AEs of interest at a frequency defined by FDA (e.g., in PSRs or on a yearly basis).

Web posting of potential safety signals

In accordance with statutory requirements and established policies and procedures, FDA has a website posting of *potential* signals of serious risks or new safety information that were identified from FAERS, or for which FAERS data were contributory [102]. A new report is made available each quarter. Listing of a drug and potential safety issue on this quarterly posting enables this information to be shared with the public at an early stage of FDA's evaluation, while simply indicating that the FDA has identified a potential safety signal for further evaluation. Therefore the FDA may not have determined or communicated what type of regulatory action, if any, is appropriate for the issue at the time of posting. Information from previous quarters is updated and the drug safety issue remains active until an FDA regulatory action has been taken or a determination is made that no regulatory action may be required. Types of regulatory action can include: modifying the drug labeling, gathering additional data to characterize the risk, development or a modification of a REMS, or suspending or withdrawing a drug from the market. In addition to the quarterly posting, a separate website posting includes signals evaluated under the Sentinel program [90]. The information posted on the Sentinel website is provided as part of FDA's commitment to make knowledge acquired from the Sentinel system readily available to the public.

Risk evaluation and mitigation strategies (REMS)

Some drugs may require additional measures to mitigate risks and ensure benefits are achieved. FDAAA amended the FD&C Act to authorize FDA to require a REMS when FDA determines that a REMS is necessary to ensure that the benefits of a drug outweigh its risks [103]. A REMS utilizes risk minimization strategies beyond the drug labeling and can provide safe access to drugs with known serious risks that would otherwise be unavailable to patients. Even after a drug is approved, new safety information may become available that requires the use of a REMS to ensure the benefits of the drug continue to outweigh the risks. An FDA guidance for drug manufacturers is available for determining when a REMS may be necessary [103]. Some of the considerations used to determine if a REMS is needed for a new safety issue include estimated size of the population using the drug, seriousness of the disease or condition being treated, expected benefit of the drug, duration of treatment, and seriousness of the AE [103].

After determining that a REMS is necessary, the goals of the REMS and specific strategies to meet the goals are considered. A REMS can include (1) a medication guide [104]; (2) a communication plan; (3) additional safety precautions, termed ETASU (elements to assure safe use); (4) an implementation system, and requires a timetable for assessment of the REMS. Medication guides are patient-friendly labeling and can be required if it could help prevent serious adverse events. A communication plan to health care providers may be required to support implementation of the REMS, to disseminate information to health care providers regarding REMS requirements and about serious risks, or to explain certain safety protocols, such as medical monitoring through periodic laboratory tests. ETASU can be required as part of a REMS to mitigate a specific serious risk listed in the drug labeling if, in the absence of an ETASU, the drug would otherwise not be approved or would be withdrawn. The following are specific elements that may be part of an ETASU: certification and specialized training of health care providers who prescribe the drug, ongoing periodic patient monitoring in order to continue drug therapy, enrollment of treated patients in a registry, or that the drug be dispensed or administered to patients with evidence or other documentation of safe use conditions [103]. Table 26.8 lists some examples of REMS components for selected drugs [105].

Manufacturers of drugs with a REMS are required to periodically assess the REMS to determine if the goals of the REMS are being met and to modify them if they are not. These assessments are submitted to FDA and reviewed by specialists. The drug manufacturer and FDA collaborate to modify the REMS throughout the life cycle of the drug as new information becomes available with expanded use.

TABLE 26.8 Examples of risk evaluation and mitigation strategies components.

Component	Drug example	Purpose
Medication guide	Qsymia (weight loss)	– Informs of congenital malformation risk and the importance of pregnancy prevention
Communication plan	Bydureon (diabetes)	– Dear Health care provider Letter informs health care providers of acute pancreatitis risk
Elements to ensure safe use	Revlimid (multiple myeloma)	– Certifies and enrolls health care providers and pharmacies in the REMS program – Pregnancy testing must occur within a certain time of prescribing – Drug dispensed to enrolled patients with evidence of safe-use conditions
Implementation system	Revlimid (multiple myeloma)	– Requires that drug manufacturer maintain a secure database of certified physicians, pharmacies, and enrolled patients and monitor compliance with elements to ensure safe use

Disclaimer

The views expressed are those of the authors and do not necessarily represent the position of, nor imply endorsement from, the US Food and Drug Administration or the US Government.

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